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Think Before You Speak: Cultivating Communication Skills of Large Language Models via Inner Monologue

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Abstract

The emergence of large language models (LLMs) further improves the capabilities of open-domain dialogue systems and can generate fluent, coherent, and diverse responses. However, LLMs still lack a crucial ability: communication skills. This limitation renders them more like information seeking tools rather than anthropomorphic chatbots. Communication skills, such as topic transition, proactively asking questions, concept guidance, empathy, and summarising often should be taken into consideration, to make LLMs more anthropomorphic and proactive during the conversation, thereby increasing the interest of users and attracting them to chat for longer. However, enabling these communication skills in black-box LLMs remains a key challenge because they do not have the same utterance formation mode as real people: think before speaking. Inspired by linguistics and cognitive science, we empower LLMs with communication skills through inner monologues. To evaluate various communication skills, we construct a benchmark named Cskills, which can also more comprehensively evaluate the dialogue generation ability of the model. Experimental results show that the proposed CSIM strategy improves the backbone models and outperforms the baselines.

1 Introduction

Open-domain dialogue systems need to generate fluent, coherent, and diverse responses based on history utterances. The emergence of large language models (Chowdhery et al., 2022; OpenAI, 2022; Touvron et al., 2023) further enhances the capabilities of dialogue generation systems and can meet the above requirements. However, LLMs are more like an information seeking tool than a chatbot like a real person. Such a dialogue system may make users lose interest in chatting and terminate the conversation. The reason is that LLMs still lack an important conversational ability: communication skills. As shown in Figure 1, LLM makes recommendations without a thorough comprehension of the preferences of the user regarding movie genres. This lack of detailed understanding may result in inaccurate recommendation outcomes. People use proactively asking questions in communication skills to further understand the needs of the user, thereby making better recommendations.

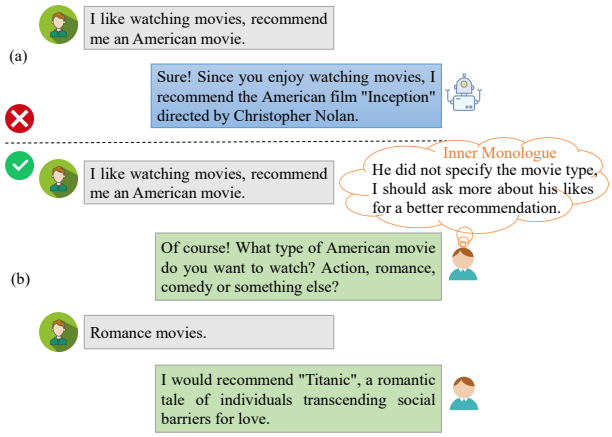

Figure 1: When asked to recommend: (a) ChatGPT directly recommends without asking the detailed needs of users, which may lead to failure to satisfy users; (b) people proactively ask questions to further understand the needs of users before making recommendations.

In linguistics, communication skills are used to enhance the interactive experience during the conversation and to establish effective communication (Dörnyei, 1995; Grover, 2005; Barker, 2010). The five common communication skills are topic transition, proactively asking questions, concept guidance, empathy, and summarising often. Each communication skill is applicable to different conversational situations and plays a different role during the conversation. By using topic transition (Dörnyei, 1995; Riou, 2015), we can avoid unfamiliar concepts and transition to familiar ones, leading to better conversations. Proactively ask-

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ing questions (Grover, 2005) can help us clarify ambiguous information and make appropriate responses based on it. Concept guidance (Zou et al., 2021) can strengthen the connection of concepts in a conversation and increase the proactivity of the dialogue. Empathy (Rizzolatti and Sinigaglia, 2008) can produce more personal and informative responses, increasing the interest of the speaker in chatting. Summarising often (Barker, 2010) allows speakers to confirm whether a consensus has been reached on the previous information, reducing the occurrence of misunderstandings.

Introducing communication skills to LLMs is not easy because they do not have the same utterance formation mode as real people: think before speaking. That is, LLMs do not have the same thinking process as real people before generating responses. LLMs are black-boxes makes understanding their decision-making process more challenging. Existing works in psychology and cognitive science indicate that humans think before speaking when they have a conversation (Hulme et al., 1999; Khawaja et al., 2008; Neustein, 2012). Li et al. (2020) proposes conversations can be decomposed into four segments: listening, thinking, speaking, and waiting, which also illustrates the importance of thinking before speaking. Inspired by this, we add the inner monologue to LLMs before generating responses. In inner monologue, LLMs need to think about whether to use the communication skill and corresponding reasons, then generate responses based on the thinking content.

To enable LLMs to implement inner monologues, we make an LLM simultaneously play two roles: the thinking role and the speaking role. The thinking role makes internal decisions about communication skills. The speaking role generates responses and chats with users. We technically use prompt engineering (Brown et al., 2020) and in-context learning (ICL) to achieve the above process. Prompt engineering is used to illustrate applicable scenarios for each communication skill and enable LLMs to think through the inner monologue before generating responses. ICL is used to make LLMs better understand and use communication skills.

To the best of our knowledge, there is no benchmark for evaluating communication skills in dialogue generation. In order to evaluate the effect of dialogue generation after adding communication skills, we constructed evaluation data for each communication skill to form a benchmark, named Cskills. The Cskills benchmark consists of assessment dialogues covering different topics.

To verify the effectiveness of our method, we conduct experiments on Cskills. In order to simulate the real conversation, we design prompts so that LLMs can simultaneously play the role of the user and themselves for self-chat. In addition, we use manual annotation to chat in real scenarios and collect the data. Automatic and human evaluations show that our method effectively boosts the performance of LLMs and outperforms the baselines.

Our contributions to this paper are three folds:

- We endow LLMs with **c**ommunication **s**kills and **i**nner **m**onologue (CSIM) through prompt engineering and in-context learning, making LLMs more anthropomorphic and proactive.
- We propose a benchmark Cskills for evaluating various communication skills, which can more comprehensively evaluate the dialogue generation ability of the model.
- We conduct comprehensive experiments on Cskills. Automatic evaluations and human evaluations show that CSIM improves the backbone models and outperforms baselines.

2 Communication Skills

In linguistics, communication skills are used to establish effective communication and increase the satisfaction of speakers during the conversation. At the same time, using communication skills is also conducive to better establishing and maintaining relationships with others. Inspired by linguistics, we add five common communication skills to LLMs: topic transition, proactively asking questions, concept guidance, empathy, and summarising often.

2.1 Topic Transition

Unfamiliar concepts and topics that do not want to be talked about should be avoided when communicating (Dörnyei, 1995). By using topic transition (Riou, 2015), we can avoid them and transition the topic to familiar or desired content, leading to better conversations. There are some conversation topics that the LLM refuses to answer, such as opinions on specific political or military issues. Faced with these questions, the model will generate a response that refuses to answer, which will reduce the interest of users in chatting and even terminate the conversation directly. In addition, part of LLMs

are forced to generate responses when facing unfamiliar topics, which may lead to wrong information in the response, misleading users, or making users feel that the other party is not a real person.

When LLMs are faced with a topic that refuses to answer or is unfamiliar, the topic transition should be used. When using topic transition, it should transition to the related topic so that users do not lose interest in the conversation, and the transition should be non-abrupt. These challenges are what LLMs need to face. Its formal definition is $P(R \mid H, t_r)$, where LLMs generate responses R by given history utterances H and the related topic t_r.

2.2 Proactively Asking Questions

When people speak, they omit certain information in certain scenarios, potentially resulting in ambiguity, such as an unclear reference. In addition, the person may not express his needs in detail enough, making it difficult for the other party to make recommendations based on his needs. LLMs ignore these ambiguities or detailed requirements of users.

Proactively asking questions (Grover, 2005) can clarify ambiguity and further understand the needs of the speaker. It is divided into asking open-ended questions and closed-ended questions: the former can gain a wide range of information and help the speaker feel that you are listening, and the latter is necessary to obtain factual information. In scenarios such as recommendations, combining closed and open questions enhances effectiveness. Its formal definition is $P(R_q \mid h, i)$, where LLMs generate responses containing the question R_q by given the ambiguous information in history utterances i and other parts of history utterances h.

2.3 Concept Guidance

Human conversations are accompanied by frequent changes of concepts, and the lack of concept management may lead to loose connections between concepts, resulting in incoherence between utterances (Zou et al., 2021). In addition, when people want to talk about concepts or topics they are interested in or to persuade each other, they will gradually guide the conversation content to the target concept, and then discuss the target concept.

Concept guidance (Zhang et al., 2019a) can better control the concept change during the conversation and strengthen the connection of concepts, and guide the conversation to the target concept. Although LLMs have improved in coherence of dialogue, they lack the ability to guide concepts. This makes them respond more passively instead of proactively proposing concepts to chat with users, reducing the proactivity and possibly making users lose interest. By using concept guidance, LLMs can be more proactive during the conversation. Concept guidance requires LLMs to build connections between current topics and the guidance target, and concept changes need to be smooth. Its formal definition is $P(R \mid H, t_s, t_g)$, where LLMs generate responses R by given history utterances H, source topic t_s and the guidance target topic t_g.

2.4 Empathy

Humans have an innate ability to form deep emotional connections with others, and empathy is at the root of complex relationships (Rizzolatti and Sinigaglia, 2008). Empathy is reflected in encouraging the other part to talk about his experience and express emotions, listening patiently and proactively responding to his utterances and emotions during the conversation (Kelley and Kelley, 2013).

Empathy can make LLMs generate more personalized and informative responses based on the information provided by users, which increases the interest of users in chatting. How to make LLMs show empathy is a challenge. Its formal definition is $P(R \mid h, p)$, where LLMs generate responses R by given the personalized information in history utterances p and other parts of history utterances h.

2.5 Summarising Often

When the rounds of the conversation increase, the information in history utterances increases accordingly, summarization is useful at this time. Summarising often (Barker, 2010) allows speakers to confirm whether a consensus has been reached on the previous information, reducing the occurrence of misunderstandings. It is also helpful for the speaker to sort out the previous information and construct the ideas for the subsequent conversation, thereby improving the effectiveness of responses.

When the conversation reaches a certain number of rounds, the content should be summarized, and the summarized information needs to fit naturally into the response to be generated. Its formal definition is $P(R \mid H_i)$, where LLMs generate responses R by given informative history utterances H_i.

3 Inner Monologue

In psychology and cognitive science, the pauses during speaking are related to the thinking pro-